# Venkata Lakshmi Parimala Pasupuleti

Data Analyst · Financial Services Data Governance · Python · SQL · AWS · Statistical Modeling

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## PROFESSIONAL SUMMARY

Data Analyst with 6+ years delivering **data governance and quality management in financial services** — enforcing compliance standards, maintaining data lineage and metadata, and building audit-ready pipelines across 1M+ financial records at production scale. At Entrata, maintained **100% PHI compliance with zero violations** over 6+ years, tuned **200+ SQL queries for 30% performance improvement**, and built automated financial data pipelines on AWS (S3, EC2, RDS, Glue). Currently at Intuit building ML pipelines (XGBoost, scikit-learn) from data ingestion through feature engineering and prediction APIs. Built multi-source statistical models achieving **$R^2 = 0.977$** across 190+ countries (Python, R) and **85%+ classification accuracy on 104K+ records**. Designed strategic BI dashboards improving decision-making comprehension **40%** (Tableau, ggplot2). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## WORK EXPERIENCE

### Intuit

Jun 2026 – Present

**Software Engineer — Data & ML Pipelines · Financial Analytics**

- Build end-to-end ML pipelines (scikit-learn, XGBoost) from data ingestion and feature engineering through model evaluation and prediction APIs for financial reporting and customer analytics workflows.
- Design structured data schemas and retrieval pipelines that ground model outputs in customer financial records — ensuring auditability and traceability required for financial reporting downstream.
- Build regression test suites (golden datasets + automated quality checks) that catch data quality regressions and model drift before production releases; ship via GitHub Actions CI/CD to AWS.
- Develop FastAPI data services and React/TypeScript dashboards surfacing analytics insights to internal stakeholders.

### Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

**Analytic Software Engineer — Income Fraud Detection · Financial Data Governance · Compliance Analytics**

- Reviewed and verified income documentation for **millions of tenant certifications** across Section 8/HCV, LIHTC, and HUD portfolios — identifying inconsistencies between stated income and supporting documents that flagged ineligible or misrepresented applications for escalation.
- Built **property performance dashboards** across 100K+ unit portfolios — eligibility approval/denial patterns, waitlist demand by bedroom size and income band, occupancy and vacancy rates, rent burden vs. AMI and voucher limits, delinquency rates, and lease renewal trends; applied LIHTC/Section 42 and HUD income-limit rules to validate unit mix, rent limits, and tenant documentation for compliance reporting across Section 8/HCV, HOME, and LIHTC programs.
- Eliminated **80% of manual eligibility review workload** with zero compliance violations by encoding the decision logic applied manually into automated Section 8/HCV, LIHTC, and HUD compliance pipelines — AMI threshold calculations, recertification workflows, and escalation routing across 100K+ unit portfolios.
- Achieved **100% PHI compliance** across all client data by migrating to AWS S3 with Libsodium-encrypted pipelines — zero security incidents; improved SQL query performance **30%** across 200+ queries supporting 1M+ property records at 99.5% uptime.

## PROJECTS

### CrashClarity: Strategic Analytics on 104K Records with Self-Service Dashboard

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble with feature engineering achieving **85%+ accuracy**; deployed interactive self-service dashboard translating model outputs into strategic severity-reduction insights for non-technical stakeholders.

### Leasel: Financial Services Data Quality & Compliance Audit Pipeline

Python · LangGraph · scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end financial data quality and governance pipeline across 1,240 documents — feature extraction, rule-based classification, and deterministic audit engine with full metadata and lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready.

### Real Estate Price Analysis: Connecticut Property Market (2001–2022)

Python · XGBoost · Random Forest · Pandas · Matplotlib · Seaborn · Feature Engineering

- Socioeconomic impact analysis of Connecticut residential property prices across two decades; XGBoost and Random Forest models on assessed value, sale ratio, property type, and geographic features surfaced key drivers of price variation across income brackets and property classes.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, data governance, ML pipelines, anomaly detection, data visualization

**B.Tech Computer Science & Engineering** — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

## TECHNICAL SKILLS

**Data Governance & Quality:** Data Quality Management, Metadata & Lineage, Audit Pipelines, PHI/HIPAA Compliance, Zero-Incident Data Migrations, LIHTC/HCV/HUD Regulatory Workflows, Access Governance, Security Controls

**Analytics & Modeling:** Python, R, SQL, PL/SQL, Statistical Modeling, Regression Analysis, Random Forest, Logistic Regression, XGBoost, Isolation Forest, Feature Engineering, Cross-Validation, scikit-learn, Pandas, NumPy

**Cloud & Data Infrastructure:** AWS (S3, EC2, RDS, Glue), ETL Pipelines, PostgreSQL, MySQL, SQLite, MongoDB, Snowflake, Docker, CI/CD (GitHub Actions), Git, Linux

**BI & Process:** Tableau, Power BI, ggplot2, Quarto, Matplotlib, Seaborn, Jupyter, Agile/Scrum (120+ sprints), FastAPI, REST APIs, TypeScript, JavaScript